TP2 - Standardized PCA: Interpretation

Results

The eigenvalues of the correlation matrix are 3.7397 (PC1) and 0.2402 (PC2). The explained variance ratio is 93.49% for PC1 and 6.01% for PC2, so the first two components keep 99.50% of the total variance.

Comparison between the two implementations

The coordinates, the eigenvalues and the explained variance ratios obtained from scratch and with scikit-learn are identical (the difference is 0.0000). The two plots of the students are the same.

Sign differences (Q17)

An eigenvector v and its opposite -v are both valid for the same eigenvalue, so the sign of a principal component is arbitrary. The signs depend on the algorithm used (np.linalg.eig for the implementation from scratch, SVD for scikit-learn). In our run the signs are the same, but they can change with another machine or library version. In that case the plot is mirrored along that axis, while the distances between students, the eigenvalues, the variance ratios and the interpretation do not change.

Interpretation

The four subjects are strongly correlated (between 0.77 and 0.99), so one component is enough to summarize most of the information. PC1 has almost equal positive weights on all subjects, so it represents the general level of a student: S4 and S2 are the best students, and S3 and S5 are the weakest. PC2 (6%) opposes Statistics and Programming: S6 is relatively better in Statistics, and S1 is relatively better in Programming.